NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Cloud Computing

Assignment – Week 9

Type of Question: MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1:

Statement: In the fog layer, latency is less but the probability of network congestion is high.
The above statement is:

a) True
b) False

Correct Answer: (b)
Detailed Solution: Both latency and network congestion is less in the fog layer

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QUESTION 2:

The term “Fog Computing” was introduced by:

a) IBM
b) Huawei
c) Cisco
d) Amazon

Correct Answer: (c)
Detailed Solution: The term "Fog Computing" was introduced by the Cisco Systems as new model to ease wireless data transfer to distributed devices in the Internet of Things (IoT) network paradigm

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QUESTION 3:

Smartwatch is an example of

a) Cloud
b) Fog node
c) Edge device
d) None of the above

Correct Answer: (c)
Detailed Solution: Refer to Cloud-Fog-Edge computing pyramid

QUESTION 4:

Involving only the cloud for directly processing the IoT applications may not be an efficient design. Which of the following could be the reason:

a) Processing time
b) Network bandwidth
c) A large amount of generated data
d) All of the above

Correct Answer: (d)
Detailed Solution: Refer to slide 1 (Challenges in Cloud Only scenario) of the lecture

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QUESTION 5:

A model that makes use of both edge resources and the cloud is referred to as:

a) Cloud computing
b) Edge computing
c) Fog computing
d) None of the above

Correct Answer: (c)
Detailed Solution: A model that makes use of both edge resources and the cloud is referred to as "fog computing"

QUESTION 6:

The _____ coordinates distributed compute nodes and performs deployment of VMs or containers to each fog/edge node

a) hardware
b) middleware
c) system software
d) none of these

Correct Answer: (b)
Detailed solution: Middleware runs on an operating system and provides complementary services that are not supported by the system software. The middleware coordinates distributed compute nodes and performs deployment of VMs or containers to each fog/edge node.

QUESTION 7:

Which of the following are not the characteristics of fog infrastructure?

a) Scalability
b) Mobility
c) Data consolidation
d) High latency

Correct Answer: (c) and (d)
Detailed solution: Fog infrastructure consists of IoT devices (End layer), Fog Nodes, and at least one Cloud Data Center (Cloud layer), with the following characteristics: location awareness and low latency, better bandwidth utilization, scalable, support for mobility

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QUESTION 8:

Which of the following type of algorithm captures the performance of resources for deciding and maximizing the performance of deployments

a) Discovery
b) Benchmarking
c) Load-balancing
d) Placement

Correct Answer: (b)
Detailed solution: Benchmarking algorithm captures the performance of resources for decision-making to maximize the performance of deployments

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QUESTION 9:

Which of the following are the benefit of cloud federation?

a) Maximize resource utilization
b) Minimize power consumption
c) Centralized processing
d) All of the above

Correct Answer: (a) and (b)
Detailed solution: Centralized processing is not aimed in the cloud federation. It is rather aimed at distributing the computation.

QUESTION 10:

Which of the following cloud architecture combines on-premise infrastructure with remote resources from one or more public clouds to provide extra capacity during peak demand?

a) Broker architecture
b) Aggregated architecture
c) Bursting architecture
d) Multitier architecture

Correct Answer: (c)
Detailed solution: Bursting or hybrid cloud architecture combines on-premise infrastructure with remote resources from one or more public clouds to provide extra capacity during peak demand

**********END**********